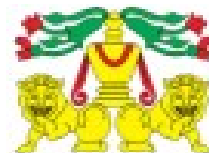




**National youth Services Council
Computer Network Technician
Final Examination
2026 January – June**



Answer all the questions. (100Marks)

Time: - 5 hours

Practical paper

VTC Network Infrastructure Upgrade

1. Project Overview

- **Organization:** VTC (IT Institute, Sri Lanka)
- **Current State:** Primary level (Peer-to-Peer) network
- **Target State:** Client-Server Domain infrastructure
- **Total Users:** 500+ Staff members

2. Core Server Roles & Services (30 Marks)

To fulfill the organizational requirements, the following Windows Server roles must be installed and configured on your lab servers:

- **Active Directory Domain Services (AD DS):** For centralized user accounts and computer management.
- **DHCP Server:** For the automatic IP address management system.
- **Windows Server Update Services (WSUS):** For centralized Windows update and patch management.
- **System Center Configuration Manager (SCCM) / GPO:** For software distribution and third-party patch management.
- **File Server Resource Manager (FSRM):** For file screening and quota management.
- **Windows Server Backup:** To configure automated backup services for all core server roles.

3. Active Directory Logical Structure & Delegation (15 Marks)

The Active Directory hierarchy must be structured with the following Organizational Units (OUs), and specific administrative rights must be delegated:

Required Departments (OUs):

- IT, Management, Admin, Lecture, HR, Operation, Marketing, Sales & Finance

Delegation of Control:

- **HR Department:** Granted explicit permissions to create new User Objects in Active Directory.
- **IT Department:** Granted explicit permissions to change/reset passwords for all domain users.

Note: To manage this, the Remote Server Administration Tools (RSAT) for AD DS must be installed on the local client accounts for the IT and HR staff so they do not need to log in to the Domain Controller directly.

4. File Server & Storage Management (FSRM) (15 Marks)

Centralized storage must be configured using FSRM with the following strict quota and screening policies:

- **Storage Quotas:**
 - **Department Shares:** Every OU must have a dedicated shared folder with a hard quota of **10GB**.
 - **User Shares:** Every individual user must have a mapped personal shared folder with a hard quota of **5GB**.
- **File Screening Policies:**
 - **Executables (.exe):** Allowed *only* in the IT OU shared folders. All other OUs and users are strictly blocked from copying .exe files.
 - **Media Files (Audio/Video):** Strictly blocked from being copied into individual user folders.

5. Group Policy Objects (GPO) (15 Marks)

The following system and security policies must be enforced via Group Policy Management:

Domain-Wide Password Policy:

- **Maximum Password Age:** 15 days.
- **Minimum Password Length:** 10 characters.
- **Complexity Requirements:** Must contain uppercase letters, lowercase letters, and at least **2 special symbols**.

Desktop Customization:

- Every OU must have a unique, department-specific desktop wallpaper applied.

System Restrictions:

- **Control Panel & IP Settings:** Restricted for all users; accessible **only to IT OU users**.
- **Command Prompt (CMD):** Restricted for all users; accessible **only to IT OU users**.
- **Software Installations:** Standard users are restricted from installing software (managed centrally).

6. Configure following group policy also. (25 Marks)

1. Every OU must have separate wallpaper.
2. Control panel and IP address settings can access only IT OU users only.
3. Every user must change password within every 15days.
4. Password complexity must be configured as bellow.
 - a. At least 10 characters must be there.
 - b. Capitol & Simple letters must be there.
 - c. At least 2 symbols should be there.
5. Restrict Software Installations
6. Restrict command prompt access to all OU's expect IT HR OU users can create Users in AD.

IT OU Users can change password in ad. ADDS must install each on their accounts.